R#1

Q1: If the argument against vapor cell includes mm-thick walls for limiting short-range interactions below 100 um, I am not understanding how cold-atom comagnetometers over come this range in part by not having 'cell walls'---isn't the cold-atom comagnetometer contained within a vacuum chamber?

Answer: In searches for short-range interactions, the distance between the test mass and the sensing atoms must be modulated. In the vapor-cell approach, in order to vary the distance, the test mass is placed outside the cell rather than being immersed in the atomic vapor. Consequently, the minimum achievable distance is limited by the cell wall thickness, typically on the millimeter scale. By contrast, in the cold-atom comagnetometer approach, both the test mass and the atoms can be placed inside the same vacuum chamber, with no walls or windows between them. Here, the atoms are held in a micrometer-scale optical trap. As a result, the distance between the test mass and the cold atoms can be reduced to the micrometer scale (well below $100\ \mu m$).

Additional details have been added in the "Introduction" section of the main text on page 2.

Q2: The total field used herein is relatively high compared to some of the other paramagnetic/paramagnetic-diamagnetic/and diag\magnetic pair systems you've mentioned in the introduction, which you allude to in the introduction. Doesn't this affect your absolute energy resolution quite significantly as compared to these systems, as a ppm on 0.1 G is not as impressive as a ppm on the much lower fields used in those studies?

Answer: In this work, the bias magnetic field does not affect the absolute energy resolution of the comagnetometer.

For a single-species ^{171}Yb magnetometer with a precession time of 10 s under a 0.1 G field, the single-shot absolute energy (frequency) sensitivity is $2\pi \times 2$ mHz. This sensitivity is primarily limited by spin-projection noise and residual light-shift fluctuations from the optical trap, while noise contributed by the 0.1 G magnetic field itself plays only a minor role.

For the ^{171}Yb - ^{173}Yb comagnetometer, the common-mode suppression factor against magnetic-field fluctuations is 3×10^4, rendering magnetic-field-induced noise negligible. Critically, sensitivity to new-physics interactions remains unsuppressed, with single-shot sensitivity maintained at $2\pi \times 2$ mHz. Accumulating data over 24 hours (~ 3000 measurements) yields a sensitivity of $2\pi \times 40\ \mu Hz$. A comparison with other relevant works is provided in the following answer to Q3.

We have added details in the "Realization of cold-atom comagnetometer" section of the main text on page 5.

Q3: Related to 2, how does your ppm performance and absolute energy resolution compare to other trapped atom systems more similar to your own?

Answer: Two previous works based on trapped atom systems are most relevant for comparison.

Gomez *et al*. [24] studied two hyperfine levels of ^{87}Rb possessing electron spins and magnetic moments. By taking a combination of the precession signals from these two levels that cancels common-mode magnetic noise on the electron-spin states, the nuclear spin precession signal was extracted. However, this construction yields a magnetometer based on the nuclear spin of ^{87}Rb rather than a comagnetometer, so it is qualitatively different from our work.

Thekkeppatt *et al*. [25] implemented a comagnetometer using ^{87}Rb and ^{87}Sr. To suppress rapid decoherence of ^{87}Sr induced by the optical-trap light shifts in their work, a magnetic field of 100–1000 G was applied – much higher than in our work. This high field, however, limited the ^{87}Rb coherence time to the millisecond scale and constrained its energy sensitivity. As reported in in their paper [25], with 1.5 h of integration, the ratio of spin precession frequencies reached a precision of 1.4 ppm, and the energy (frequency) sensitivity was $2\pi \times 0.12$ Hz. This 1.4 ppm precision was limited by systematic errors and remained unchanged with longer integration times, making it comparable to our 4 ppm result. On the other hand, their energy sensitivity would improve with longer integration; for example, reaching $2\pi \times 25$ mHz over 24 h.

For comparison, the energy sensitivity of our comagnetometer is $2\pi \times 40\ \mu\text{Hz}$ over 24 h, which represents an advantage of more than two orders of magnitude. The key to this advantage is our investigation and control of light shifts in the optical trap.

Q4: The proposals at the end of the paper involving EDM searches, including those that incorporate 223Ra are 225Ra are ambitious, but I wonder how much these actually have to do with your current manuscript? You give no hard estimates of improvements sensitivity of your proposals herein, so it does not make sense to me to include them here.

Answer: Following the Reviewer comment, we have removed the text on the outlook of the Ra case. Please see page 10 in the main text.

R#2

Q1: The "cat" state used for Yb-173, is it what is called a NooN state? Should the connection be mentioned?

Answer: The so-called NOON state, which takes the form $|N,0\rangle + |0,N\rangle$, describes a coherent superposition of N particles all occupying one quantum state and all occupying another orthogonal state. The spin cat state used in our work, $|F,+F\rangle + |F,-F\rangle$, exhibits an analogous structure in Hilbert space. Following the Reviewer comment, we have added a corresponding mention in the main text, in section "Realization of cold-atom comagnetometer" on page 4, and cited *J. Opt. Soc. Am. B* **27**, A25–A29 (2010) [28].

Q2: In Fig. 4, there are significant deviations of the experiment from the model, why?

Answer: The model used to simulate the data in Fig. 4 was oversimplified. In this measurement, ^{173}Yb atoms were prepared in a coherent spin state (not a cat state) encompassing all m_F states, and these states were not fully resolved in the spin-state readout. However, the original model accounted only for the detection of $m_F = \pm 5/2$ states. Moreover, it assumed a uniform value for the tensor light shifts across the optical trap. Consequently, the simulation exhibited sharp peaks absent from the experimental data.

Following the Reviewer comment, we have developed a more realistic model and performed new simulations. The revised model incorporates the different detunings and the Clebsch–Gordan coefficients（see Fig. 5b）for transitions from different m_F states. Specifically, when intending to probe the $|m_F = +5/2\rangle$ state, the detection probabilities for all m_F states are: 1.00, 0.58, 0.31, 0.16, 0.08, and 0.03 (from +5/2 to −5/2). Additionally, the revised model includes the distribution of tensor light shifts across the optical trap. As shown in the updated Fig. 4, the sharp features present in the previous simulation are now smoothed, leading to improved agreement with the experimental data. Details of the revised simulation are provided in the Methods section, under "Calibration of Ω_B/Ω_T".

It should be noted that the detection issue described above is specific to the coherent spin state used for calibration. In contrast, the cat states prepared elsewhere in the experiment involve only $|m_F = \pm 5/2\rangle$ states, which have maximally different detection probabilities and are thus clearly resolvable. Details are provided in the Methods section, under "Atom sample detection".

Q3: Please check the formatting of the references. I noticed capitalization problems in Refs. 19 and 26.

Answer: Following the Reviewer comment, we have corrected the capitalization errors throughout the reference list. These included, for instance, the change of "yb" to "Yb" in Ref. 26 and similar corrections in Refs. 11 and 23.

Q4: The authors mention the simultaneous effect of the Zeeman and tensor light shifts. In fact, this is a major effect (often referred to alignment-to-orientation conversion) that plays an important role in nonlinear Faraday rotation. The authors might wish to make this connection, if they think it is relevant.

Answer: The alignment-to-orientation conversion (AOC) in nonlinear Faraday rotation stems from tensor light shifts of the probe beam. This effect induces coherent evolution of spin states, which is precisely what we employ to drive the atoms from an initial coherent spin state (orientation) to a cat state (alignment).

Furthermore, the AOC effect is also closely related to the simultaneous effect of the Zeeman and tensor light shifts in our experiment. When the laser polarization is perfectly aligned with the magnetic field, the cat state (alignment) is an eigenstate of the tensor operator. However, when a misalignment exists between the laser polarization and the magnetic field, the cat state experiences

a twist due to the tensor shift, analogous to AOC.

Following the Reviewer comment, we have addressed this connection in the "Tensor shift" section of the main text, and added a corresponding citation *Phys. Rev. Lett.* **85**, 2088–2091 (2000) [30].

R#3

The biggest achievement of the paper is, to us, the implementation of a method to suppress light shifts induced by the trapping lattice, and the subsequent measurement "with a precision of 4 ppm" of the magnetic moment ratio between 171Yb and 173Yb.

Q1a: How valuable is this measurement/Why is it important?

Answer: Magnetic moment is a fundamental property of the nucleus. The ratio of nuclear magnetic moments between different isotopes reveals the relative evolution of their nuclear structures. In order to determine an unknown nuclear magnetic moment, a widely adopted experimental approach is to measure its ratio to a known reference moment. Our measurement of the ^{171}Yb / ^{173}Yb magnetic moment ratio with 4 ppm precision demonstrates the viability of optical-trap comagnetometry for high-precision nuclear property measurements.

Q1b: Why not provide the measured value directly in the abstract?

Answer: Following the Reviewer comment, we have added the measured value to the abstract.

Q1c: Have there been more measurements of this ratio apart from these of Refs. [29] and [30]? If so, a summary plot showing each measured value and associated uncertainty would be welcome. Are all the quoted values (authors and previous works) given with a 1-σ uncertainty?

Answer: There are two previous measurements. Following the Reviewer comment, we have added Table 1 in the main text, listing both prior results alongside our result. Values in parentheses are 1σ uncertainties.

All three results are consistent within uncertainties. Our result is more precise than the previous ones by one and two orders of magnitude, respectively. Given the large differences in the error bar lengths, we find it difficult to present all three results on a single plot. Instead, we have chosen to present the comparison in a table on page 9.

Q1d: In the part Ratio of nuclear magnetic moments (page 8), the authors compile values of the ratio of nuclear magnetic moment, RL, measured while different experimental parameters are varied to search for systematic effects (Figs. 2 and 3). Why are the different values not overlapping? Please comment on how the average is weighted and why.

Answer: As listed in the table below, the four R_L values given in Figs. 2 and 3, as well as the final average value, are consistent within the quoted uncertainties.

	R_L
Fig. 2b	0.726079(3)
Fig. 2d	0.726080(5)
Fig. 3a	0.726077(2)
Fig. 3b	0.726072(4)
Final value	0.726076(3)

In Figs. 2b and 2d, R_L at the extremum point of a parabolic fit to the data is obtained under magnetic fields of 50 mG and 100 mG, respectively. This was intended to verify the validity of the tensor shift theory. The fitting results confirm that, when the tensor field is aligned with the magnetic field, R_L is independent of the magnetic field strength.

After completing the field alignment procedure, measurements were conducted at 100 mG (Fig. 3a and 3b) to investigate the dependence of R_L on key experimental parameters. No dependence was observed. Fig. 3a and 3b each yields a weighted mean value of R_L. The final R_L value is the weighted average of these two results.

Additional details have been added in the "Ratio of nuclear magnetic moments" section of the main text on page 8.

Q2a: The perspective of the paper concerns the search the of EDM using their experimental apparatus. As we understand, the main point here is that using a comagnetometer (rather than a single specie) should help reduced magnetic field noises which are currently the limiting factor.

In the introduction, authors claim that the glass vapor cell comagnetometer are unsuitable to search for "short-range interactions below ~ 100 μm". Could they elaborate? A citation would help here.

Answer: In *Phys. Rev. Lett.* **128**, 231803 (2022) [13], the authors noted that, "its (the cell) end wall perpendicular to the long axis has a thickness of 0.5 mm", and "results are mainly limited by the Rb magnetometer sensitivity, and by the minimum distance between the BGO mass and Xe atoms."

In vapor-cell experiments, the wall thickness prevents the atom–mass separation from reaching the sub-100 μm range. Moreover, atoms are distributed throughout the cell, and only wander into the region closest to the wall and the test mass for a small fraction of the time, which further dilutes

the sensitivity to short-range interactions. We have added this description in the "Introduction" section of the main text on page 2.

Q2b: Authors estimate an improvement of the sensitivity to spin-mass coupling by up to three orders of magnitude. Could they also elaborate on this? A plot showing current limits together with their projected sensitivity would, in my opinion, strengthen the paper.

Answer: The figure below illustrates the projected sensitivity to spin-mass couplings. The green curve (5) corresponds to a comagnetometer consisting of a single ^{171}Yb and a single ^{173}Yb atom held in an optical tweezer, with a minimum separation of $5\ \mu\text{m}$ between the atoms and a platinum test mass. Assuming a measurement cycle of 30 s (interrogation time: 10 s) and a total data accumulation time of 30 days (~ 30,000 cycles), the projected sensitivity to $g_s g_p$ at $\lambda = 4\ \mu\text{m}$ would improve by three orders of magnitude over the previous best limit (*Phys. Rev. D* **92**, 114001 (2015) [36]). Following the Reviewer comment, we have added this description to the main text on page 10.

We have chosen not to include the figure in the present paper. A search experiment based on these principles is currently underway. The physical parameters described above and the associated systematic errors remain under investigation. Significant additional work beyond the scope of this manuscript is required before the sensitivity plot below can be considered ready for publication.

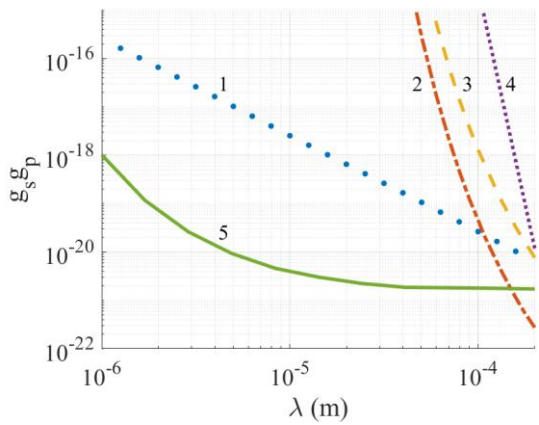

Figure 1. Constraints on the spin-mass coupling constants $g_s g_p$ as a function of the force range λ.
Line 1, ^{3}He depolarization: *Phys. Rev. D* **92**, 114001 (2015) [36].
Line 2, ^{129}Xe– ^{131}Xe–Rb comagnetometer: *Phys. Rev. Lett.* **128**, 231803 (2022) [13].
Line 3, ^{129}Xe– ^{131}Xe–Rb comagnetometer: *Phys. Rev. Lett.* **111**, 102001 (2013) [12].
Line 4, ^{3}He– ^{129}Xe comagnetometer: *Phys. Rev. Lett.* **111**, 100801 (2013) [19].
Line 5: projected sensitivity of a future search based on a ^{171}Yb – ^{173}Yb cold-atom comagnetometer as reported in this work.

Q3: "The ratio of their Larmor frequencies —termed the comagnetometer ratio—" We are not sure that this is a standard term.

Answer: The term "comagnetometer ratio" was introduced in this manuscript for convenience and is not a standard term. Following the Reviewer comment, it has been changed to "Larmor-frequency ratio" throughout the manuscript.

Q4: In the introduction, the authors mention that they have identified a "novel systematic effect stemming from the interplay between Zeeman and tensor light shifts". Does this refer to the discussion in section "Tensor shift"?

Answer: Yes. We have inserted "(see section 'Tensor shift')" in Introduction on page 3 to make this connection clear.